Punctured Curves

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Symplectizations, Symplectic Cobordisms, and Moduli Spaces

Goal:

- Define the moduli space of holomorphic curves used in SFT.

Outline:

① Stable Hamiltonian Structures p. 1-5

② Collars and Cobordisms p. 5-8

③ Symplectizations, Energy, and Completions p. 8-13

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Contact Dynamics ①

Let (M, ξ) be a contact manifold with contact form α. There exists a unique vector field R on M such that

$$\begin{cases} \iota_R d\alpha \equiv 0 \\ \iota_R \alpha \equiv 1 \end{cases}$$

R is called the Reeb vector field determined by α.

Lemma

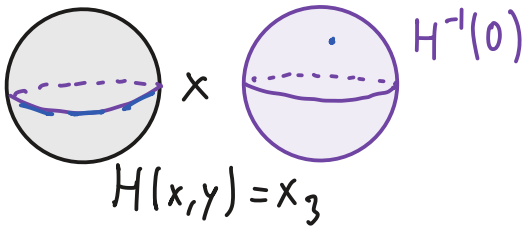
Let φ_t be the flow of R. Then,

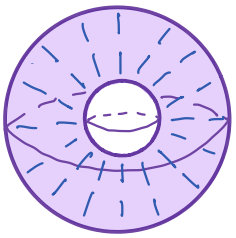
$$\varphi_t^* \alpha = \alpha \quad \text{and} \quad (\varphi_t)_* \xi = \xi.$$

Proof

We have $\frac{d}{dt}(\varphi_t^* \alpha) = \mathcal{L}_R \alpha = d\iota_R \alpha + \iota_R d\alpha = 0$, so $\varphi_t^* \alpha = \varphi_0^* \alpha = \alpha$. Since $\xi = \ker \alpha$, $(\varphi_t)_* \xi = \xi$ follows. ∎

Relatedly, let (W, ω) be a symplectic manifold, $H: W \to \mathbb{R}$ a smooth function and $M := H^{-1}(\text{regular point})$ be a smooth hypersurface in W.





The restriction $\omega_M := \omega|_{TM}$ is a closed 2-form of maximal rank on M. Its kernel is the characteristic line field, whose integral curves are orbits of the Hamiltonian vector field generated by H.

In these two examples, ω_M and $d\alpha$ behave very similarly. This similarity is captured by the following definition:

Definition

A Hamiltonian structure on a smooth $(2n-1)$-manifold M is a closed 2-form ω with maximal rank. The 1D distribution $\ell_\omega := \ker \omega_M$ is called the characteristic line field of ω.

If M is oriented, ℓ_ω inherits an orientation. We call a positively oriented nowhere zero section $X \in \Gamma(\ell_\omega)$ a Hamiltonian vector field on (M, ω). The space of such X is an open and convex subset of $\Gamma(\ell_\omega)$.

We have generalized $d\alpha$, but what about α itself?

Definition

> Given an oriented manifold M with a Hamiltonian structure ω, a framing of ω is a choice of 1-form λ such that λ is positive on ℓ_ω. We call (ω, λ) a framed Hamiltonian structure.

The space of framings of ω is nonempty and convex.

Just like in the contact case, there is a unique Hamiltonian vector field R such that

$$\begin{cases} \iota_R \omega \equiv 0 \\ \iota_R \lambda \equiv 1 \end{cases}.$$

Moreover, λ determines a complementary vector bundle for ℓ_ω:

$$\xi := \ker \lambda \subset TM.$$

Note that $\omega|_\xi$ is nondegenerate, giving $\xi \to M$ the structure of a symplectic vector bundle.

Let φ_t be the flow of R.

Good: $\varphi_t^* \omega = \omega$.

Sad: $(\varphi_t)_* \xi \neq \xi$ in general.

Let's fix this!

Definition

A framing λ of a Hamiltonian structure ω on M is called stable if $\iota_R d\lambda \equiv 0$. We call (ω, λ) a stable Hamiltonian structure (SHS).

$$\text{Good: } (\varphi_t)_* \xi = \xi \quad \text{if } \lambda \text{ stable.}$$

That φ_t preserves ω immediately earns us the following result:

Proposition

Suppose (ω, λ) is a framed Hamiltonian structure on M with Reeb vector field R, flow φ_t, and $\gamma : (a,b) \to M$ a solution to $\dot{\gamma} = R \circ \gamma$. Then, for any $t_0, t_1 \in (a,b)$ the linear map

$$\Pi_\xi \circ (d\varphi_{t_1 - t_0})_{\gamma(t_0)} : \xi_{\gamma(t_0)} \longrightarrow \xi_{\gamma(t_1)}$$

is a symplectic isomorphism. If λ is stable, we can omit Π_ξ above.

We get a unique symplectic connection ∇^ω on $\gamma^* \xi$ such that parallel transport along γ is given by composing Π_ξ with the linearized Reeb flow.

Fact: If ∇ is any symmetric connection on M, then

$$\nabla^\omega_t \eta = \Pi_\xi (\nabla_t \eta - \nabla_\eta R)$$

If λ is stable, we can omit Π_ξ above.

We generalized the contact case to SHS's. But, why bother doing this if we will focus on the contact case anyway?

Cool Application: Hamiltonian Floer is a special case of SFT. If (W, Ω) is a closed symplectic manifold, $H: S^1 \times W \to \mathbb{R}$ smooth function. Then, its symplectic gradient X_t defines a symplectic connection on the trivial symplectic fiber bundle

$$M := S^1 \times W \to S^1.$$

Setting $\omega = \Omega + dt \wedge dH$ and $\lambda = dt$ makes (ω, λ) a SHS on M. The closed orbits of its Reeb vector field correspond to 1-periodic Hamiltonian orbits on W and we go from there ...

Hypersurfaces of Symplectic Manifolds ②

As motivation for Hamiltonian structures, we saw any hypersurface of a symplectic manifold inherits a Hamiltonian structure. Given a framing of the structure, we can recover the symplectic structure of a neighborhood of M:

Proposition

Suppose M is a smooth oriented hypersurface in a symplectic manifold (W, ω) and associate to any given vector field $V \in \Gamma(TW|_M)$ along M the 1-form

$$\lambda := \iota_V \omega|_{TM} \in \Omega^1(M).$$

Then, V is positively transverse to M if and only if λ is a framing of the Hamiltonian structure $\omega_M = \omega|_{TM}$. Moreover, if this holds, M is compact and contained in the interior of W, then a neighborhood $\mathcal{N}(M) \subset W$ of M admits a

symplectomorphism

$$(N(M), \omega) \cong ((-\varepsilon, \varepsilon)_r \times M, \omega_M + d(r\lambda))$$

identifying $M \subset N(M)$ with $\{0\} \times M$, and V with ∂_r.

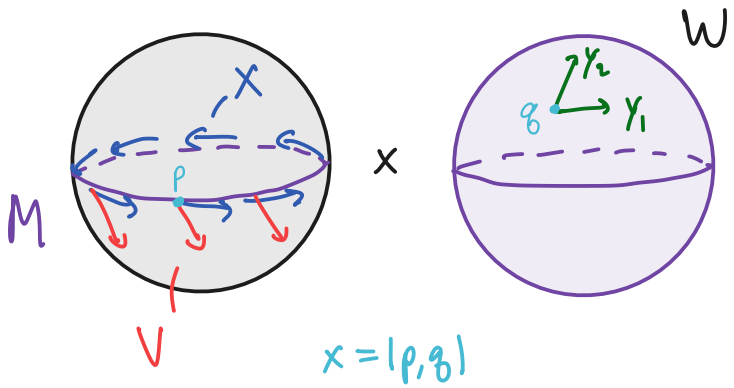
Proof Sketch

Let X be a Hamiltonian vector field. We have to check that V is positively transverse if and only if $\lambda(X) > 0$. Fix $x \in M$. If $V \in TM_x$, then $X \in \ker \omega_M$ implies $\lambda(X)(x) = 0$. Otherwise, pick a positively oriented basis of $\xi_x = \ker \lambda_x$, $Y_1, \dots, Y_{2n-2}$. It follows

$$\omega^n(V, X, Y_1, \dots, Y_{2n-2}) = C\omega(V, X)\omega^{n-1}(Y_1, \dots, Y_{2n-2}),$$

where $C > 0$ and $\omega^{n-1}(Y_1, \dots, Y_{2n-2}) > 0$ by assumption. Hence, $\lambda(X) = \omega(V, X)$ and $\omega^n(V, X, Y_1, \dots, Y_{2n-2})$ have the same sign, as desired.

Now, assume $\lambda = \iota_V \omega|_{TM}$ is a framing. Extending V arbitrarily to a neighborhood of M, we may use its flow φ_t^V to obtain an embedding



$$(-\varepsilon, \varepsilon) \times M \hookrightarrow W; \ (r, x) \mapsto \varphi_r^V(x)$$

for $\varepsilon > 0$ sufficiently small.

This identification already maps $\{0\} \times M$ to M and ∂_r to V, but we do not know that $\omega = \omega_0 := \omega_M + d(r\lambda)$.

Fact: There is a 1-form β on $(-\varepsilon, \varepsilon) \times M$ such that

$$\omega - \omega_0 = d\beta \quad \text{and} \quad \beta|_M = 0.$$

We can now modify our embedding using the Moser trick:

Let $\omega_t = \omega_0 + t\,d\beta$ and pick Y_t the unique solution to

$$\iota_{Y_t}\omega_t = -\beta.$$

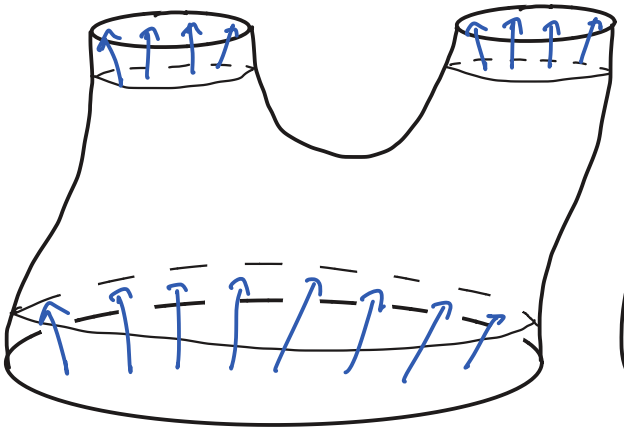
Let ψ_t be the flow of Y_t. Note that $Y_t = 0$ along M, so on a possibly smaller neighborhood ψ_1 is well-defined. Moreover, ψ_t fixes M. Finally,

$$\frac{d}{dt}\left(\psi_t^*\omega_t\right)\Big|_{t=a} = \psi_a^*\left(\mathcal{L}_{V_t}\omega_t + \frac{d\omega_t}{dt}\right) = \psi_a^*\left(d\iota_{V_t}\omega_t + d\beta\right) = 0,$$

so $\psi_1^*\omega = \psi_1^*\omega_1 = \psi_0^*\omega_0 = \omega_0$. ∎

Remark: In the contact case, it suffices to just flow along V. The Moser modification is not necessary.

If M is a boundary component of W instead of lying in the interior we obtain analogous statements with a collar neighborhood. If V points outward, we obtain a collar of the form $(-\varepsilon, 0] \times M$. If V points inward, we get $[0, \varepsilon) \times M$.



Definition

Given two closed $(2n-1)$-dimensional oriented manifolds $M_\pm$ with Hamiltonian structures $\omega_\pm$, a symplectic cobordism from (M_-,ω_-) to (M_+,ω_+) is a compact symplectic $2n$-manifold (W,ω) whose boundary has an orientation-preserving diffeomorphism to $-M_- \sqcup M_+$ identifying $\omega|_{T(\partial W)}$ with $\omega_\pm$ on $M_\pm$.

If the Hamiltonian structures have framings $\lambda_\pm$, we abbreviate $\mathcal{H}_\pm := (\omega_\pm, \lambda_\pm)$ and refer to (M,ω) as a symplectic cobordism from $(M_-,\mathcal{H}_-)$ to $(M_+,\mathcal{H}_+)$.

If the framings are stable, we say the symplectic cobordism has stable boundary.

We refer to M_+ (M_-) as the positive (negative) boundary.

This labeling is a choice that can be freely reversed. We just have to be careful that if we have framings, we have to replace $\lambda \to -\lambda$.

What SFT cares about is the form of the collar neighborhoods $N(M_\pm) \subset W$.

Cylindrical Ends ③

In the case of a contact manifold (M,ξ), there is a canonical symplectic structure on $\mathbb{R}\times M$, called the symplectization of M.



If M is endowed with a framed Hamiltonian structure $\mathcal{H} = (\omega, \lambda)$ instead, there is no single canonical symplectic structure on $\mathbb{R} \times M$, but the collar neighborhoods from before help us:

Fix $\varepsilon > 0$ small enough for $\omega + d(r\lambda)$ to be symplectic on $(1-\varepsilon, \varepsilon) \times M$. Define

$$\mathcal{T} := \left\{ \varphi \in C^\infty(\mathbb{R}, (-\varepsilon, \varepsilon)) \mid \varphi' > 0 \right\}.$$

Then, any $\varphi \in \mathcal{T}$ gives a symplectic form $\omega_\varphi := \omega + d(\varphi(r)\lambda)$.

We want to give $\mathbb{R} \times M$ an almost complex structure that is in some sense compatible with $\mathcal{H}$.

Definition

Given a framed Hamiltonian structure $\mathcal{H} = (\omega, \lambda)$ with Reeb vector field R and hyperplane distribution ξ, denote by

$$\mathcal{J}(\mathcal{H}) \subset \mathcal{J}(\mathbb{R}_r \times M)$$

the space of almost complex structures J on $\mathbb{R} \times M$ satisfying

- J is invariant under the $\mathbb{R}$-action on $\mathbb{R} \times M$ by translation.
- $J\partial_r = R$ and $JR = -\partial_r$.
- $J(\xi) = \xi$ and $J|_\xi$ is compatible with $\omega|_\xi$.

When the framing is stable, all these almost complex structures will satisfy a tameness condition.

We say J is <u>tamed by</u> $\mathcal{H}$ if we can choose $\varepsilon > 0$ such that ω_φ tames J for every $\varphi \in \mathcal{T}$.

Proposition

The following are equivalent:

(1) Every $J \in \mathcal{J}(H)$ is tamed by H.

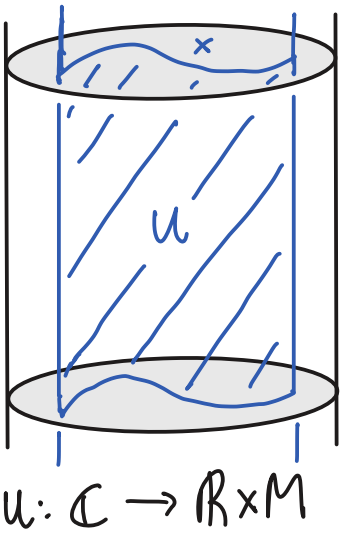
(2) There exists $J \in \mathcal{J}(H)$ that is tamed by H.

(3) The framing λ is stable.

In this setting, we define the energy of a J-holomorphic curve $u:(\Sigma, j) \to (\mathbb{R}\times M, J)$ by

$$E(u) = \sup_{\varphi \in \Gamma} \int_\Sigma u^* \omega_\varphi .$$

Tamedness implies $E(u) \geq 0$ with equality if and only if u is constant.



Given any orbit $x:\mathbb{R}\to M$ of R, the curve $u:\mathbb{C}\to\mathbb{R}\times M$ mapping $s+it \mapsto (s, x(t))$ is J-holomorphic.

Bad: u has infinite energy.

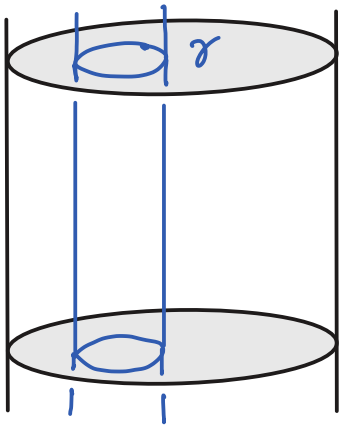
Example: $M = S^1$, $\omega = 0$, $\lambda = dt$, $x(t) = t$.
Then, $u: \mathbb{C} \to \mathbb{R}\times S^1$; $s+it \mapsto (s,t)$.

Identify $\psi: \mathbb{R}\times S^1 \to \mathbb{C}\setminus\{0\}$; $(s,t) \mapsto e^{2\pi(s+it)}$ biholomorphically.
Then, $\psi \circ u : \mathbb{C} \to \mathbb{C}\setminus\{0\}$ is $z \mapsto e^{2\pi z}$ which has an essential singularity at ∞.

If $x:\mathbb{R}\to M$ is a periodic Reeb orbit with period $T>0$, we can fix this problem:

Parametrize x as $\gamma: S^1 \to M$; $t \mapsto x(tT)$ so $\dot{\gamma} = T \cdot R(\gamma)$. Let

$$u_\gamma : \mathbb{R} \times S^1 \longrightarrow \mathbb{R} \times M$$
$$(s, t) \mapsto (Ts, \gamma(t))$$



Then, u_γ is J-holomorphic for any $J \in \mathcal{J}(H)$ and is called the trivial cylinder over γ.

Its energy is

$$E(u_\gamma) = \sup_{\varphi \in \mathcal{T}} \int_{\mathbb{R}\times S^1} u_\gamma^* w_\varphi = \sup_{\varphi \in \mathcal{T}} \int_{\mathbb{R}\times S^1} u_\gamma^* \, d(\varphi(t)\lambda) = 2\varepsilon T.$$

These trivial cylinders are very well-behaved. The map $u_\gamma : \mathbb{R} \times S^1 \to \mathbb{R} \times M$ is proper and its composition with the projection $\pi : \mathbb{R} \times M \twoheadrightarrow M$ converges asymptotically to a loop near each of the punctures in $\mathbb{R} \times S^1 \cong S^2 \setminus \{0, \infty\}$.

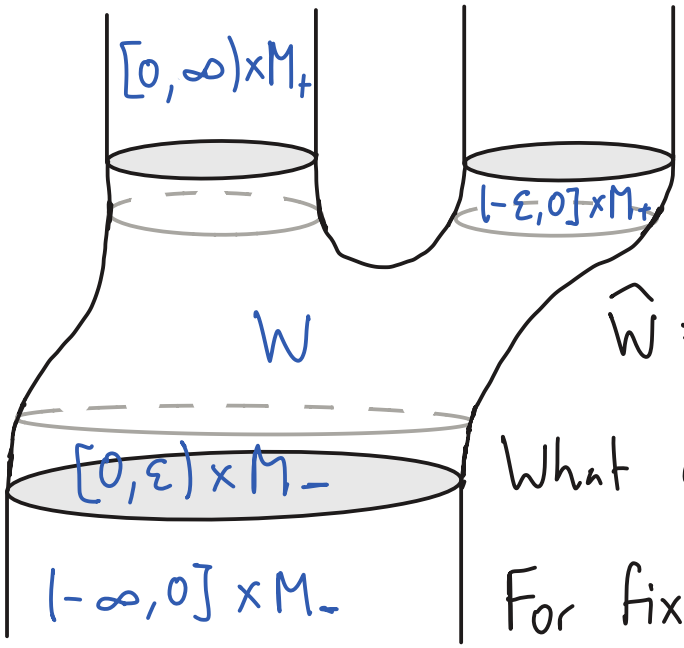
Under generic assumptions, these two properties will hold for <u>all</u> punctured holomorphic curves.

With this nice model in mind, we can apply these ideas to symplectic cobordisms:

Assume (W, ω) is a symplectic cobordism from $(M_-, \mathcal{H}_-)$ to $(M_+, \mathcal{H}_+)$, with $\mathcal{H}_\pm = (\omega_\pm, \lambda_\pm)$ framed Hamiltonian structures.

<u>Sad</u>: Curves that touch the boundary.

Fix: Get rid of the boundary!



As a smooth manifold, the completion of W is

$$\widehat{W} = ((-\infty,0]\times M_-) \underset{M_-}{\cup} W \underset{M_+}{\cup} ([0,\infty)\times M_+).$$

What about the symplectic structure?

For fixed $\varepsilon>0$ sufficiently small, let

$$\mathcal{T}_0 := \{\varphi \in C^\infty(\mathbb{R},(-\varepsilon,\varepsilon)) \mid \varphi'>0, \varphi(r)=r \text{ near } 0\}$$

Any $\varphi \in \mathcal{T}_0$ defines a symplectic form on $\widehat{W}$ by

$$\omega_\varphi := \begin{cases} d(\varphi(r)\lambda_+)+\omega_+ & \text{on } [0,\infty)\times M_+ \\ \omega & \text{on } W \\ d(\varphi(r)\lambda_-)+\omega_- & \text{on } (-\infty,0]\times M_- \end{cases}.$$

We need almost complex structures on $\widehat{W}$ to be well-behaved at infinity.

Definition

Let $\mathcal{J}_\tau(\omega,\mathcal{H}_+,\mathcal{H}_-) \subset \mathcal{J}(\widehat{W})$ denote the space of almost complex structures J on $\widehat{W}$ such that

- J on $[0,\infty)\times M_+$ matches an element of $\mathcal{J}(\mathcal{H}^+)$
- J on $(-\infty,0]\times M_-$ matches an element of $\mathcal{J}(\mathcal{H}^-)$
- J on W is tamed by ω.

Let $\mathcal{J}(\omega, \mathcal{H}_+, \mathcal{H}_-) \subset \mathcal{J}_\tau(\omega, \mathcal{H}_+, \mathcal{H}_-)$ be the subset where J is also compatible with ω on W.

If the structures $\mathcal{H}_\pm$ are both stable, then any $J \in \mathcal{J}_\tau(\omega, \mathcal{H}_+, \mathcal{H}_-)$ is tamed by ω_φ for every $\varphi \in \mathcal{T}_0$.

In this case, just as before, we define the energy of a J-holomorphic curve $u: (\Sigma, j) \to (\hat{W}, J)$ by

$$E(u) := \sup_{\varphi \in \mathcal{T}_0} \int_\Sigma u^* \omega_\varphi .$$

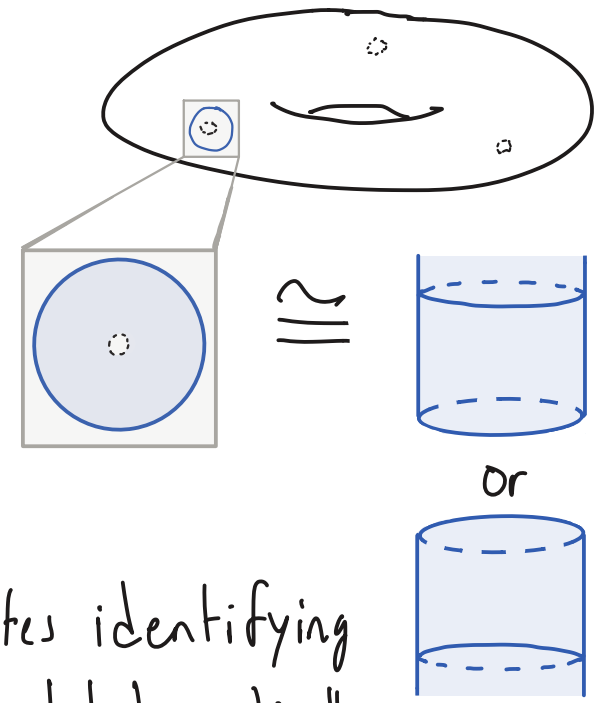
Asymptotically Cylindrical Curves

We already saw trivial cylinders are very well-behaved. One of Hofer's insights was that we can more generally consider curves on arbitrary punctured Riemann surfaces.

A punctured Riemann surface is

$$\dot{\Sigma} := \Sigma - \Gamma,$$

where (Σ, j) is a closed connected Riemann surface and $\Gamma \subset \Sigma$ is a finite set of punctures.



For any $z \in \Gamma$, we can find coordinates identifying some punctured neighborhood of z biholomorphically with the closed punctured disk

$$\dot{\mathbb{D}} := \mathbb{D} - \{0\} \subset \mathbb{C}.$$

We can identify this with either the positive or negative half-cylinder

$$Z_+ := [0,\infty) \times S^1, \quad Z_- := (-\infty, 0] \times S^1$$

$$(s,t) \mapsto e^{-2\pi(s+it)}, \quad (s,t) \mapsto e^{2\pi(s+it)}.$$

Such a choice is a (positive/negative) holomorphic cylindrical coordinate system near ζ. Note that these (arbitrary) choices partition $\Gamma = \Gamma^+ \sqcup \Gamma^-$ into positive and negative punctures.

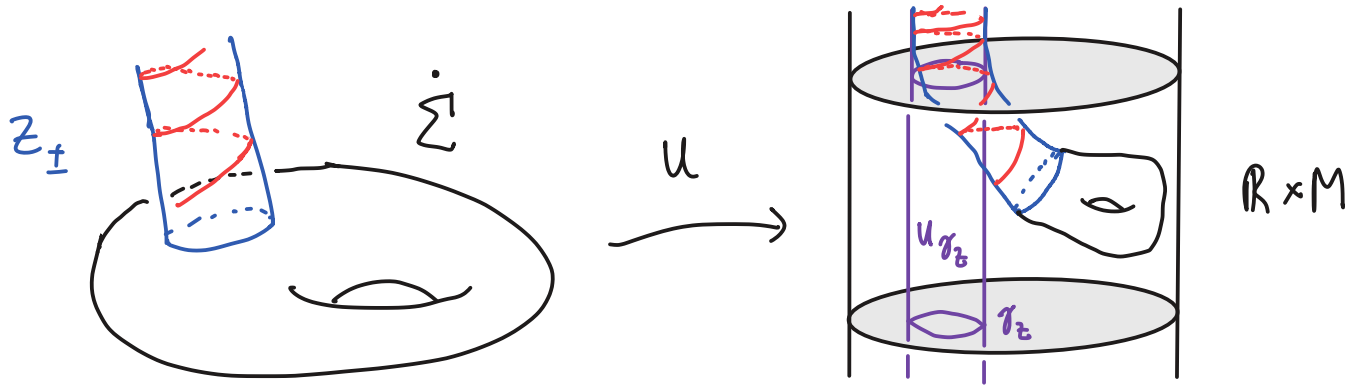
Definition

A smooth map $u: \dot{\Sigma} \to \mathbb{R} \times M$ is called asymptotically cylindrical if for each $z \in \Gamma^\pm$, there exists a closed Reeb orbit $\gamma_z : S^1 \to M$ with associated trivial cylinder $u_{\gamma_z} : \mathbb{R} \times S^1 \to \mathbb{R} \times M$, and constants $s_0 \in \mathbb{R}$ and $t_0 \in S^1$ such that

$$u(s - s_0, t - t_0) = \exp_{u_{\gamma_z}(s,t)} h_z(s,t) \quad \text{for } (s,t) \in Z_\pm \text{ with } |s| \gg 0,$$

where $h_z(s,t)$ is a vector field along u_{γ_z} satisfing

$$h_z(_ + s, _) \to 0 \quad \text{in } C^\infty(Z_\pm) \quad \text{as } s \to \pm\infty$$

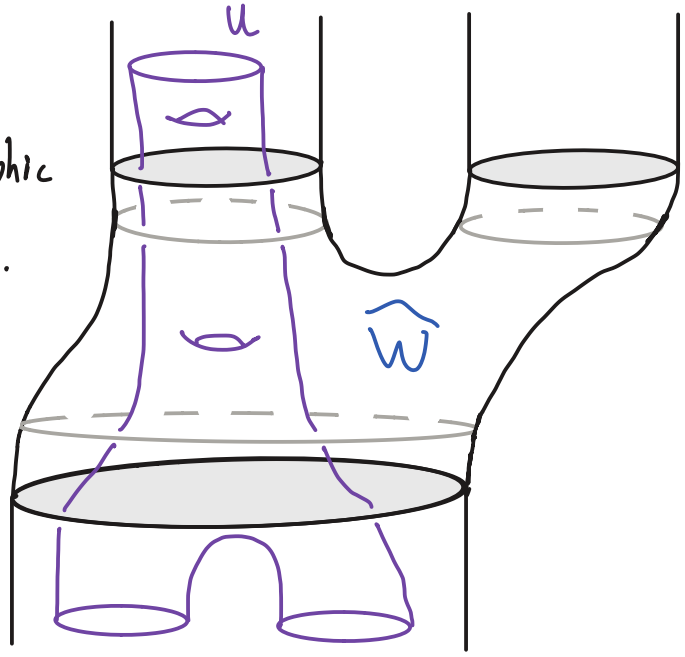


We call γ_z the asymptotic orbit of u at the puncture z, and call h_z the asymptotic representative of u at z.

Remark: This notion does not depend on the choices of holomorphic cylindrical coordinates near z. Moreover, these coordinates and u uniquely determine h_z as well as s_0 and t_0.

We can extend this to completed symplectic cobordisms. Let (W, ω) be a symplectic cobordism from $(M_-, \mathcal{H}_-)$ to $(M_+, \mathcal{H}_+)$ and $J \in \mathcal{J}_\tau(W, \mathcal{H}_+, \mathcal{H}_-)$. Denote by $\xi_\pm$ and $R_\pm$ the hyperplane distribution and Reeb vector field determined by the SHS's $\mathcal{H}_\pm = (\omega_\pm, \lambda_\pm)$.

An asymptotically cylindrical curve $u : (\dot{\Sigma}, j) \to (\widehat{W}, J)$ is a proper map that sends neighborhoods of positive/negative punctures to the positive/negative cylindrical ends of $\widehat{W}$, where they asymptotically approach trivial cylinders over closed orbits of $R_\pm$.



Asymptotically cylindrical J-holomorphic curves always have finite energy. If all Reeb orbits are nondegenerate, the converse is also true.

To each asymptotically cylindrical $u : \dot{\Sigma} \to \widehat{W}$ we can associate a relative homology class.

Let $\bar{\gamma}^{\pm} \subset M_{\pm}$ denote the closed 1-dimensional submanifold defined as the union over $z \in \Gamma^{\pm}$ of the images of the orbits γ_z. Let $\bar{\Sigma}$ denote $\dot{\Sigma}$ with $\{\pm\infty\} \times S^1$ appended to each of its cylindrical ends, and let $\bar{W}$ denote $\hat{W}$ with $\{\pm\infty\} \times M_{\pm}$ attached to its cylindrical ends. Then, u has a unique extension $\bar{u} : (\bar{\Sigma}, \partial\bar{\Sigma}) \to (\bar{W}, \bar{\gamma}^+ \cup \bar{\gamma}^-)$ and thus represents

$$[u] := \bar{u}_*[\bar{\Sigma}] \in H_2(\bar{W}, \bar{\gamma}^+ \cup \bar{\gamma}^-) = H_2(W, \bar{\gamma}^+ \cup \bar{\gamma}^-).$$

If we consider a symplectization $\mathbb{R} \times M$ instead of $\hat{W}$, we get

$$[u] \in H_2([-\infty, \infty] \times M, \bar{\gamma}^+ \cup \bar{\gamma}^-) = H_2(M, \bar{\gamma}^+ \cup \bar{\gamma}^-).$$

We can finally define the moduli space.

Fix integers $g, k_+, k_- \geq 0$ along with ordered sets of Reeb orbits $\gamma^{\pm} = (\gamma_1^{\pm}, \dots, \gamma_{k_\pm}^{\pm})$, where $\gamma_i^{\pm}$ is a closed orbit of $R_{\pm}$ in $M_{\pm}$. Denote the union of the images of the $\gamma_i^{\pm}$ by $\bar{\gamma}^{\pm} \subset M_{\pm}$ and choose a relative homology class $A \in H_2(W, \bar{\gamma}^+ \cup \bar{\gamma}^-)$ that satisfies

$$\partial A = \sum_{i=1}^{k_+} [\gamma_i^+] - \sum_{i=1}^{k_-} [\gamma_i^-] \in H_1(\bar{\gamma}^+ \cup \bar{\gamma}^-).$$

The <u>moduli space</u> of <u>unparametrized</u> J-holomorphic curves of genus g, homologous to A, and asymptotic to (γ^+, γ^-) is the set of equivalence classes of tuples

$$\mathcal{M}_{g,0}(J, A, \gamma^+, \gamma^-) = \{(\Sigma, j, \Gamma^+, \Gamma^-, u)\} / \sim$$

where

(1) (Σ, j) is a closed Riemann surface of genus g;

(2) $\Gamma^+ = (z_1^+, \dots, z_{K_+}^+)$ and $\Gamma^- = (z_1^-, \dots, z_{K_-}^-)$ are disjoint ordered sets of points in Σ;

(3) $u : (\Sigma - (\Gamma^+ \cup \Gamma^-), j) \to (\widehat{W}, J)$ is an asymptotically cylindrical J-holomorphic curve with $[u] = A$, asymptotic at $z_i^\pm \in \Gamma^\pm$ to $\gamma_i^\pm$ for $i = 1, \dots, K_\pm$;

(4) $(\Sigma_0, j_0, \Gamma_0^+, \Gamma_0^-, u_0) \sim (\Sigma_1, j_1, \Gamma_1^+, \Gamma_1^-, u_1)$ if and only if there exists a biholomorphic map $\psi : (\Sigma_0, j_0) \to (\Sigma_1, j_1)$ taking $\Gamma_0^\pm$ to $\Gamma_1^\pm$ with order preserved such that $u_1 \circ \psi = u_0$.

When there is no need to specify A, we abbreviate

$$\mathcal{M}_{g,0}(J, \gamma^+, \gamma^-) := \bigcup_A \mathcal{M}_{g,0}(J, A, \gamma^+, \gamma^-).$$

We often write u to mean $[(\Sigma, j, \Gamma^+, \Gamma^-, u)]$.

Remark: We can topologize $\mathcal{M}_{g,0}(J, \gamma^+, \gamma^-)$, but in practice we will focus on cases in which $\mathcal{M}_{g,0}(J, \gamma^+, \gamma^-)$ has some manifold/orbifold structure, so we will not bother.

Proposition

> For each choice of the data $g, A, \Gamma^\pm$, the energy $E(u)$ satisfies a uniform bound for all $u \in \mathcal{M}_{g,0}(J, A, \gamma^+, \gamma^-)$.

The Automorphism Group ⑤

The automorphism group $\mathrm{Aut}(u)$ is the group of biholomorphic maps $\psi:(\dot\Sigma,j)\to(\dot\Sigma,j)$ which act as the identity on $\Gamma^+\cup\Gamma^-$ and satisfy $u=u\circ\psi$.

We will show $\mathrm{Aut}(u)$ is always finite.

Theorem

Assume $u:(\dot\Sigma,j)\to(\hat W,\hat J)$ is a nonconstant asymptotically cylindrical $\hat J$-holomorphic curve whose asymptotic orbits are all nondegenerate. Then, there exists a factorization $u=v\circ\varphi$, where

- $\varphi:(\dot\Sigma,j)\to(\dot\Sigma',j')$ is a holomorphic map of positive degree;
- $v:(\dot\Sigma',j')\to(\hat W,\hat J)$ is an asymptotically cylindrical $\hat J$-holomorphic curve which is embedded except at a finite set of non-immersed points and self-intersections.

If φ is a diffeomorphism, we call u simple, otherwise u is a k-fold multiple cover of v, with $k:=\deg\varphi\geq 2$.

Proposition

If $[(\dot\Sigma,j,\Gamma^+,\Gamma^-,u)]\in\mathcal{M}_{g,0}(\hat J,A,\gamma^+,\gamma^-)$ is represented by a simple curve, then $\mathrm{Aut}(u)$ is trivial. If it is represented by a k-fold multiple cover of a simple curve, then $|\mathrm{Aut}(u)|\leq k$.

In particular, $\mathrm{Aut}(u)$ is always finite.

The Dream: $\mathcal{M}_{g,0}(\hat J,A,\gamma^+,\gamma^-)$ is a smooth manifold of dimension $(n-3)(2-2g-k_+-k_-)+2c_1^\tau(A)+\sum_{i=1}^{k_+}\mu_{CZ}^\tau(\gamma_i^+)-\sum_{i=1}^{k_-}\mu_{CZ}^\tau(\gamma_i^-)$.